

Gender bias in ecosystem restoration: from science to practice

Cruz-Alonso V^{1,2}, Velado-Alonso, E³, Martínez-Baroja, L^{4,5}; Marqués, L^{6,7}; Rodríguez-Uña, A⁸; Rohrer, Z; Monteagudo, N⁵

veronica.cral@gmail.com
@Veruitter

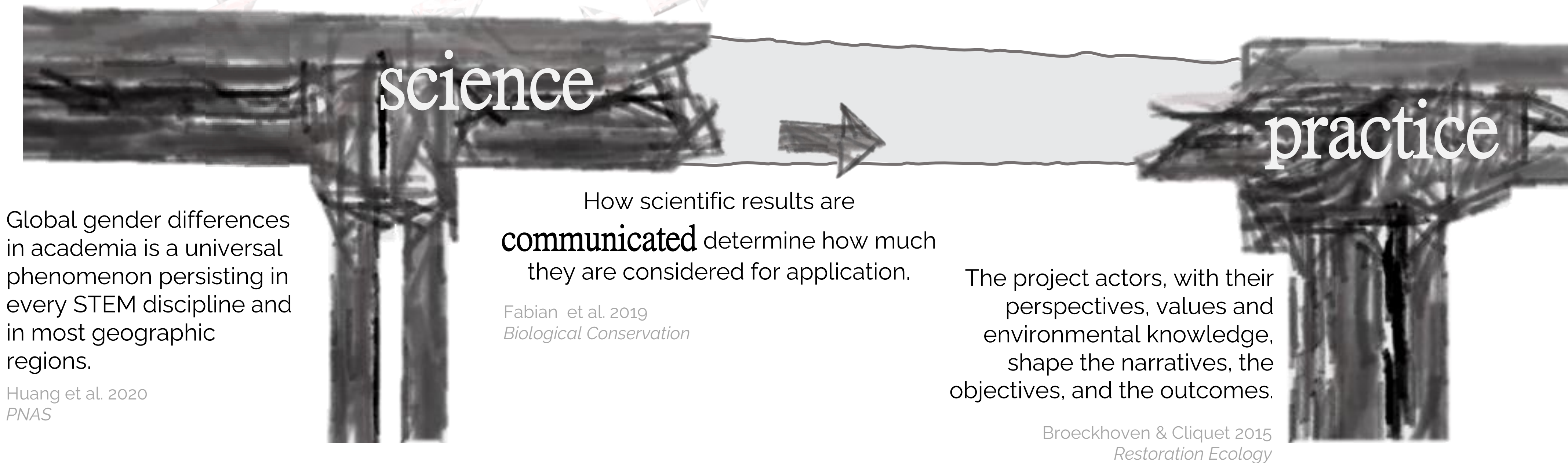
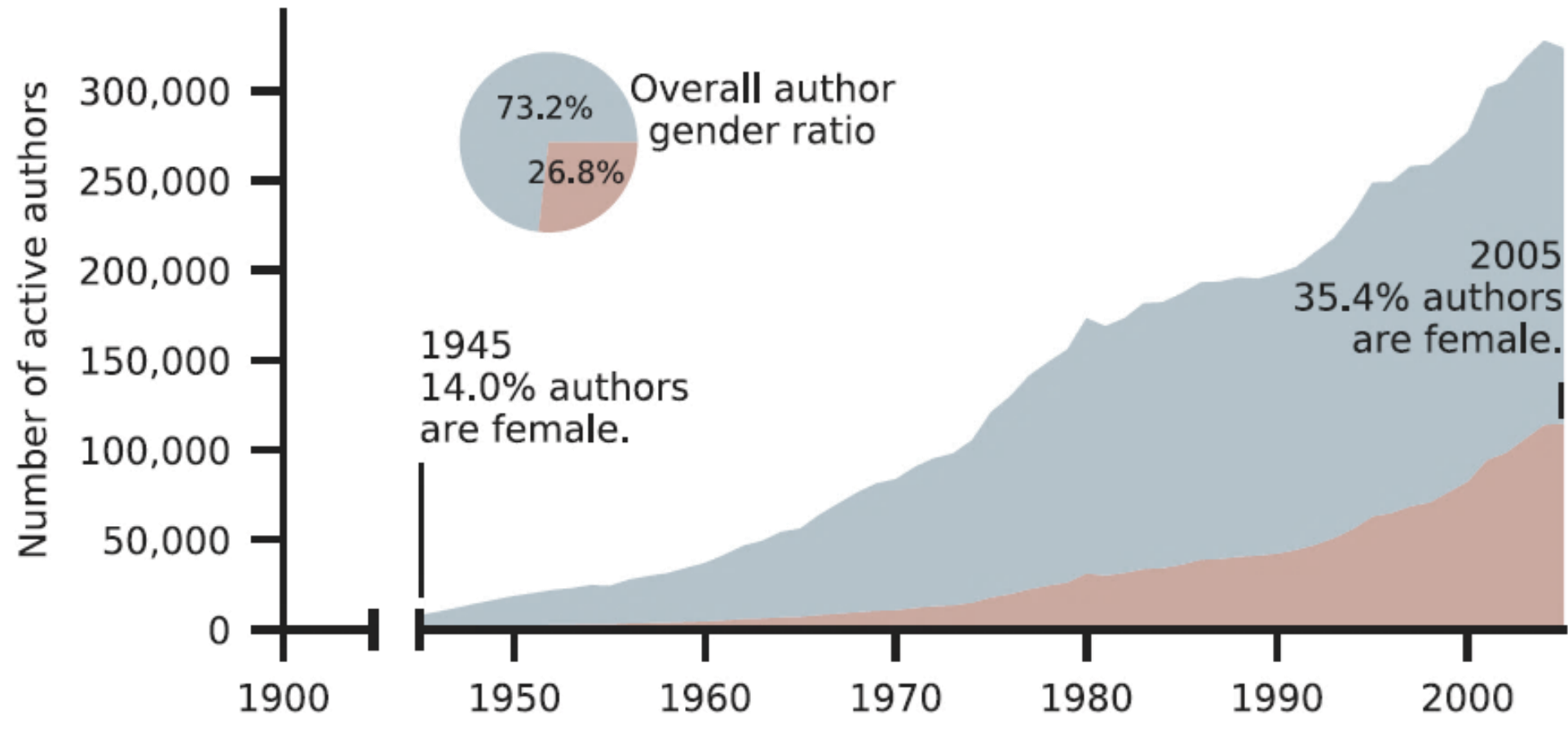
Full article
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Restoration Ecology



The declaration of the Decade on Ecosystem Restoration aims to create a **new trajectory** for the relationships between societies and nature including human rights with a focus on **gender equality**, youth, local communities, indigenous people, and future generations.

Ecosystem restoration is not gender neutral. Gender equality can be part of every restoration-related initiative (policies, plans, international agreements, research, communication, education programs, project design and implementation) both as an *outcome* and as an *intrinsic feature*.

This study focuses on the gender features of teams performing restoration initiatives, specifically the team leaders that steer restoration initiatives in research, communication, and practice.

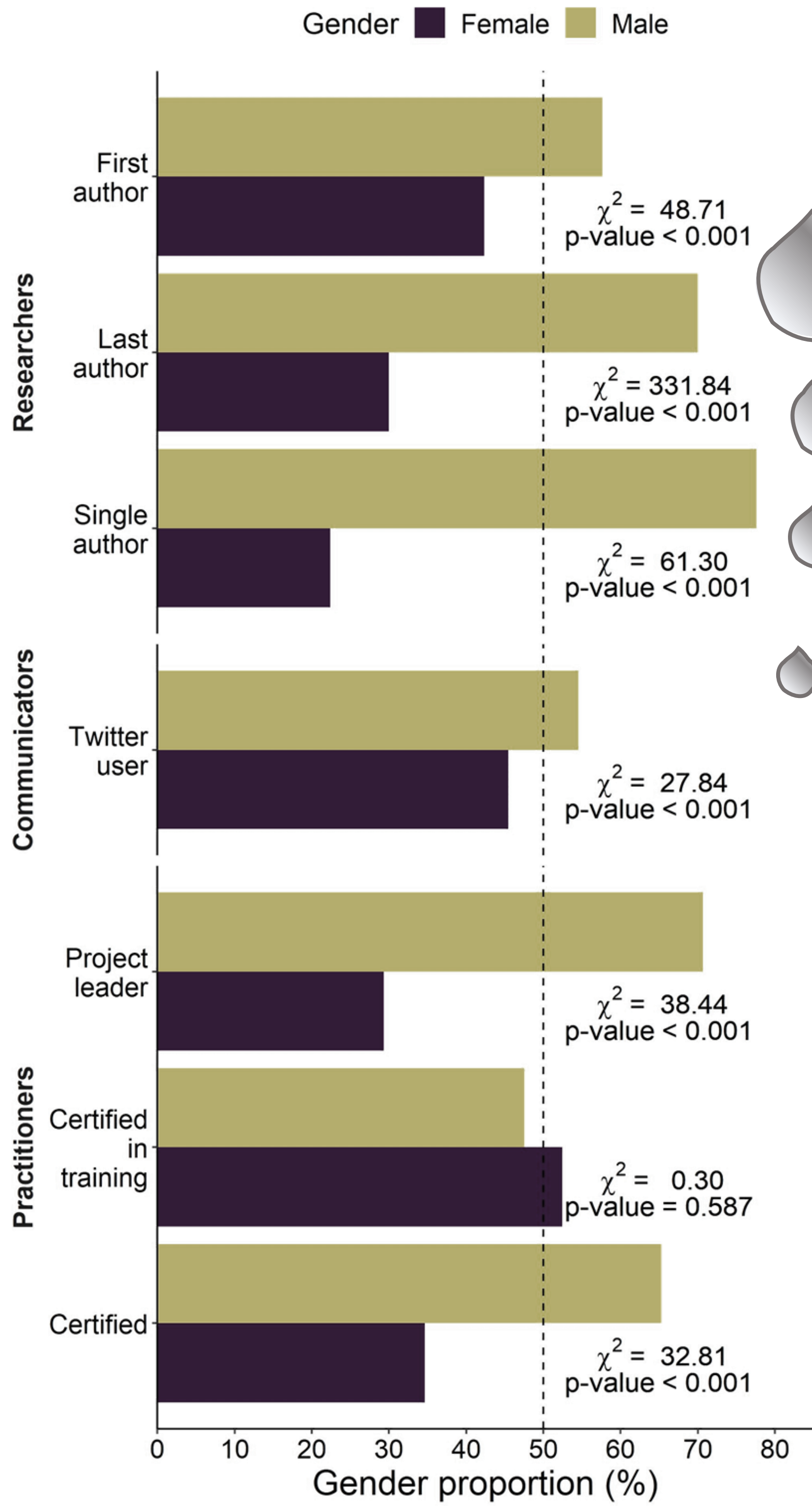


We answered **three specific questions**:

- 1) Is **gender bias** present among **people leading** ecosystem restoration initiatives in research, outreach and practice?
- 2) What are the **factors** that could influence these patterns of gender bias in ecosystem restoration?
- 3) Does the **impact of scientific knowledge** on society depend on the gender of the scientific team?

Main results

Men were the primary **leaders** in research, outreach and practice initiatives in ecosystem restoration.



Gender inequality persists within research, outreach and practice in ecosystem restoration, mainly in senior positions. These results can serve as a baseline for the design of equal restoration initiatives triggered by the UN Decade on Ecosystem Restoration.

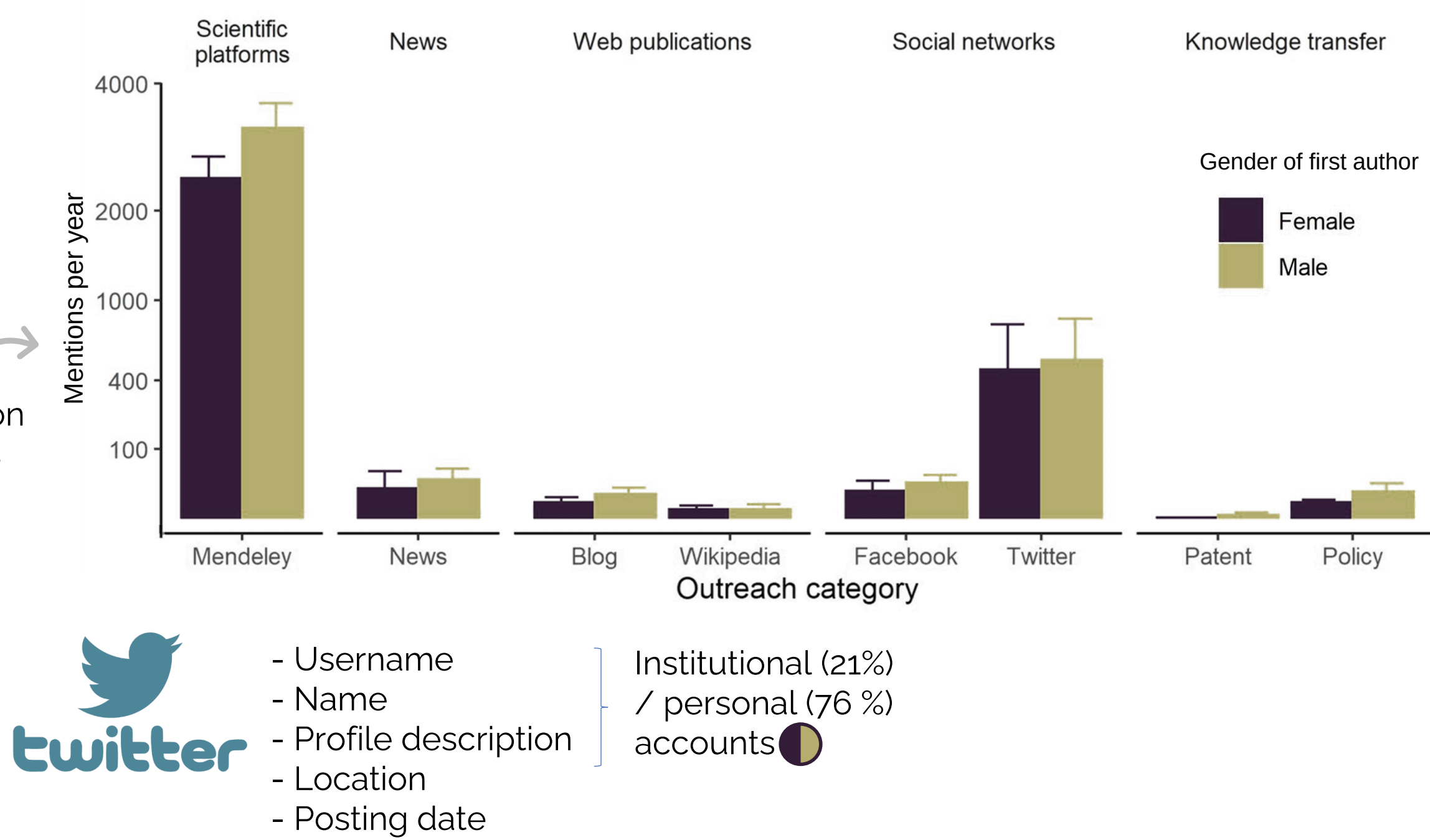
This broad perspective of inequality can evolve toward gender equality...

Methods

We used **five databases** to extract information about the leaders of restoration initiatives.

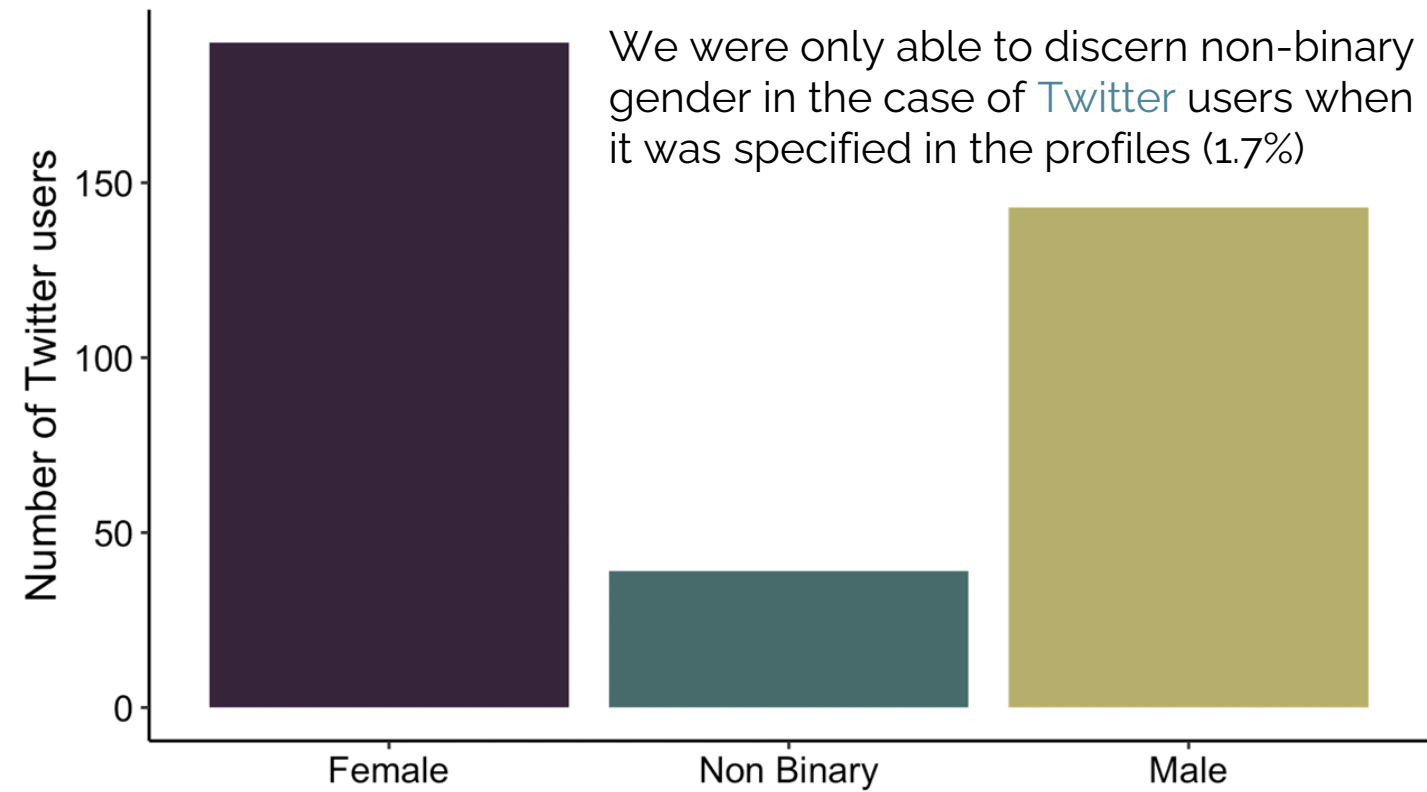
SER SOCIETY FOR ECOLOGICAL RESTORATION
SER's Project Database
- Project leader
- Affiliations
- Target ecosystem
- Year

Altmetric
- Altmetric Attention Score (AAS) and its components
- Tweet ID
- Certified practitioner
- Program type (CERP/CERPIT)

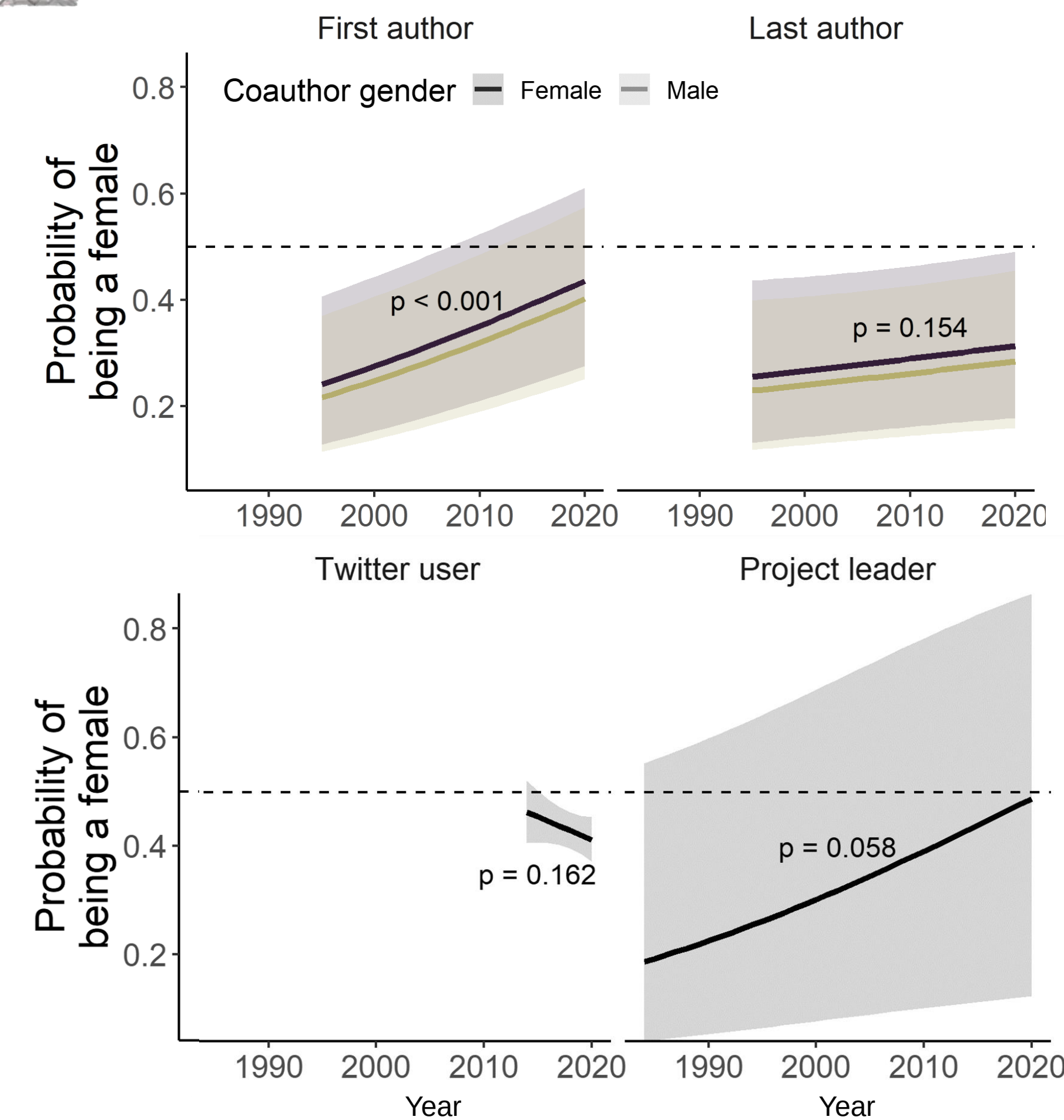


We defined the **gender** using GenderGuesser R package. Caddigan 2021

Gender describes the socially-constructed characteristics typically associated with male and female identities. We inferred people's gender based on names, pictures and other information adhering to the traditional binarism (i.e. female or male), but we are aware of the limits of this procedure regarding important gender aspects such as inferring the gender of a person from a single attribute or ignoring other realities as non-binary or trans identities.



The probability of **being a female** researcher or project leader **increased over time**, although **it was not enough to reach equality** (i.e. probability = 0.5).



When analyzing by **target ecosystem**, articles, tweets, and projects related to **forest and freshwater ecosystems** were significantly **less led by women**. Additionally, the proportion of women as last authors was significantly lower in all ecosystem types except for urban.

Projects tended to be **led by women** more frequently in **more developed countries** (higher IHDI). However, **gender bias persisted** in restoration science and practice **in all countries**.

Authors' **gender** did not affect the scientific impact of the articles (no. of citations), nor the no. of **Twitter mentions** but, when the last author was a **male researcher**, articles had **higher AAS**.

Institutions mentioned both genders equally on Twitter, but male Twitter users tended to mention articles authored by men, and female Twitter users tended to mention articles authored by women.

